Technology Stack

Date	29 June 2026
Team ID	N/A
Project Name	Rising Waters
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side**	*HTML, CSS, JavaScript, Bootstrap*	*Creates a responsive and user-friendly interface to display flood predictions, weather updates, and alerts on desktop and mobile devices.*

2	**Backend / Server-Side**	*Python, Flask*	*Processes weather and river-level data, executes the AI/ML prediction model, and serves prediction results through the web application.*
3	**Database / Data Storage**	*MySQL*	*Stores user details, historical flood records, weather data, and prediction results efficiently for future analysis.*
4	**Cloud / Hosting / Deployment**	*Render / PythonAnywhere*	*Provides cost-effective online deployment with easy access, scalability, and reliable hosting for the application.*
5	**Version Control & CI/CD**	*Git & GitHub*	*Enables version control, team collaboration, code backup, and project management throughout development.*

6	**Third-Party APIs / Other Tools**	*OpenWeatherMap API, Google Maps API, Scikit-learn, Pandas, NumPy, Matplotlib*	*OpenWeatherMap provides real-time weather data, Google Maps visualizes flood-prone areas, and Python libraries are used for data preprocessing, machine learning, and visualization.*